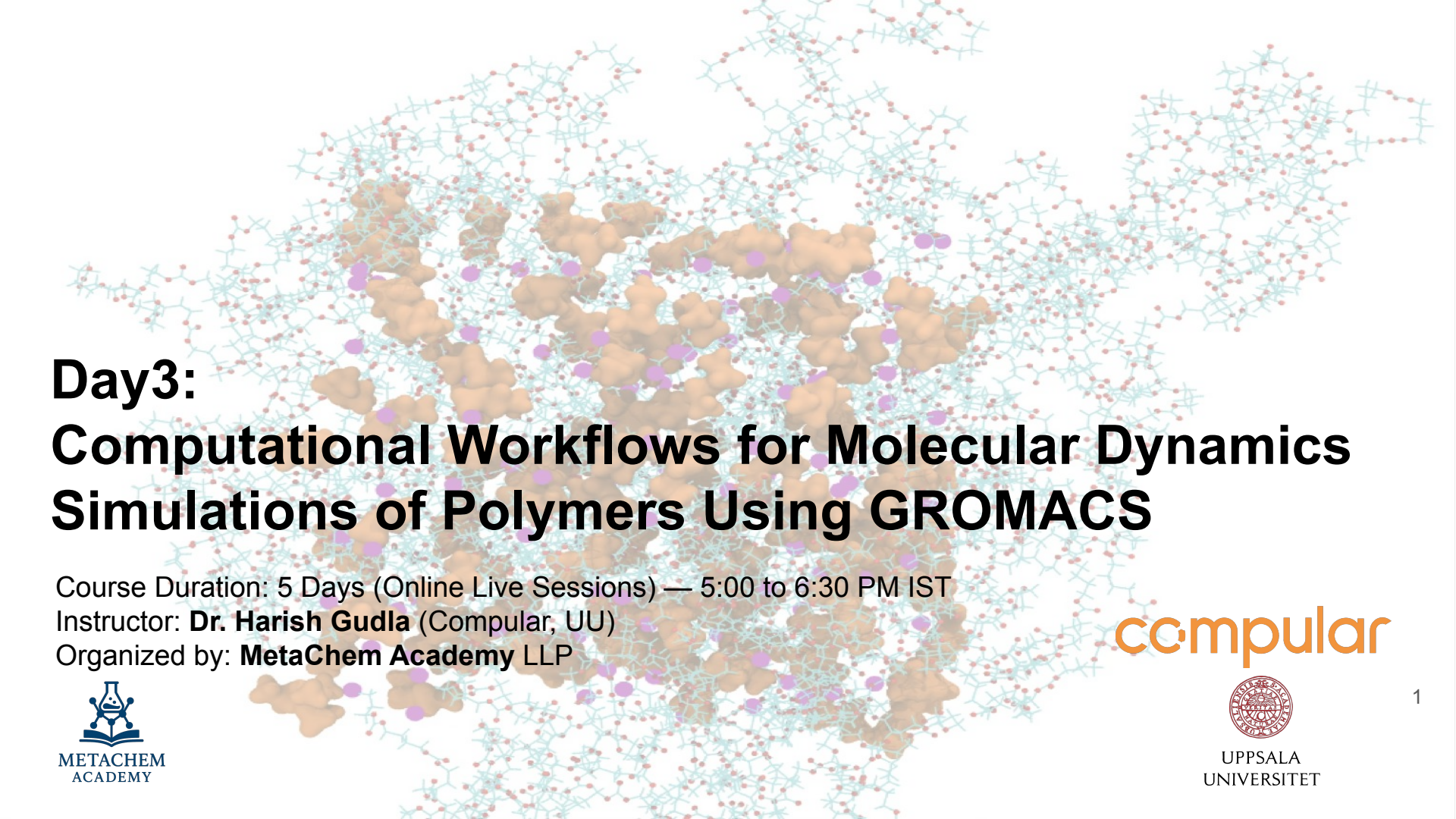




Day3: Computational Workflows for Molecular Dynamics Simulations of Polymers Using GROMACS

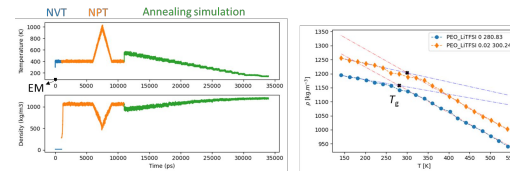
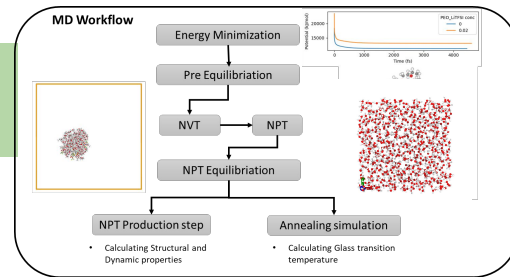
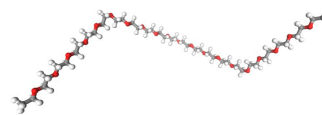
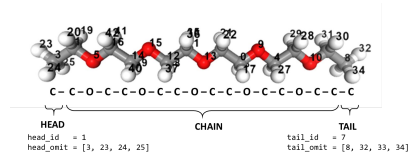
Course Duration: 5 Days (Online Live Sessions) — 5:00 to 6:30 PM IST

Instructor: **Dr. Harish Gudla** (Compular, UU)

Organized by: **MetaChem Academy LLP**

Workshop Schedule

Date	Theory Session (5:00–5:30 PM IST)	Hands-On Session (5:30–6:30 PM IST)
10 Dec	Introduction to classical MD, force fields, and motivation for polymer simulations	Installing and setting up tools and Parameterizing monomers
11 Dec	Polymer chain architecture: HEAD/CHAIN/TAIL atom definitions and polymerization logic	Getting to jupyter notebooks and Parameterizing monomers Building polymer chains using parameterized monomers and preparing input files
12 Dec	Polymer chain architecture: HEAD/CHAIN/TAIL atom definitions and polymerization logic GROMACS input formats	Building polymer chains using parameterized monomers and preparing input files Run first GROMACS MD on single polymer chain
15 Dec	Polymer simulation examples (with properties) and MD simulation workflows	Running equilibration simulations and analyzing density for multi-polymer chain systems
16 Dec	Thermal and conformational analysis: annealing simulations and glass transition temperature (Tg) concepts	Performing annealing runs and calculating polymer properties: density, RMSD, Rg, and Tg



Day 3 - Outline

- Recap
- Theory session (5:00-5:30 pm IST)
 - Polymer chain architecture
 - GROMACS input/output formats
- Hands-on session (5:30-6:00 pm IST)
 - Working with python/jupyter notebook
 - Building polymer chains using parameterized monomers
 - GROMACS MD on single polymer chain

MD algorithm

Step 1: initial positions $r^{(i=0)}$ and velocities $v^{(i=0)}$
for atoms, set $a = 0$, $t = 0$, $i = 0$, timestep Δt

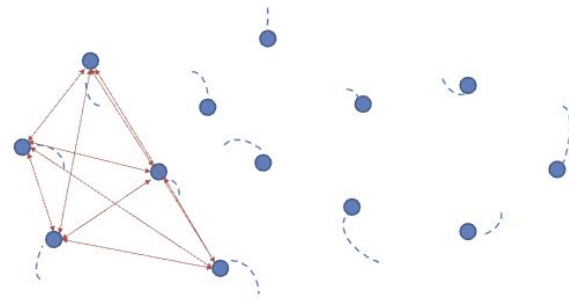
Step 2: after atom movements, location update: $r^p = r^i + v^{(i)}\Delta t + \frac{1}{2}a\Delta t^2 + \text{accurate terms}$, velocity update:
 $v^p = v^{(i)} + a\Delta t + \text{accurate terms}$

Step 3: force/energy calculations: $F = -\nabla v(r^p)$ or $F = F(\Psi(r^p))$ and $a = F / m$

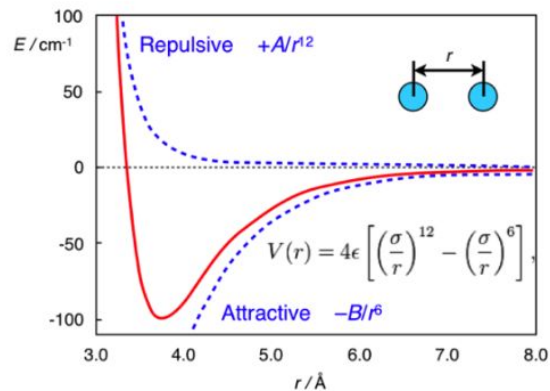
Step 4: after atom movements, location update: $r^{i+1} = r^p + f(a, \Delta t)$, velocity update: $v^{(i+1)} = v^p + f(a, \Delta t)$

Step 5: Apply boundary conditions, temperature and pressure control. Time integration and iteration step: $t = t + \Delta t$, $i = i + 1$. Repeat step 2 until the acceptable results are obtained.

N atoms



Interatomic pair potentials: $U_{1,2}, U_{1,3}, U_{1,4}, U_{1,5}, U_{2,3}, U_{2,4}, U_{2,5}, \dots, U_{N,N-1}$



Classical Force Field

The functional form of a typical forcefield

$$U_{\text{total}} = U_{\text{LJ}} + U_{\text{Coul}} + U_{\text{Bond}} + U_{\text{Angle}} + U_{\text{Dihedral}} + U_{\text{Improper}}:$$

Non-bonded interactions

$$U_{\text{LJ}} = 4\epsilon_{ij} \left[\left(\frac{\sigma_{ij}}{r_{ij}} \right)^{12} - \left(\frac{\sigma_{ij}}{r_{ij}} \right)^6 \right],$$

$$U_{\text{Coul}} = \frac{q_i q_j}{4\pi\epsilon_1\epsilon_0 r_{ij}},$$

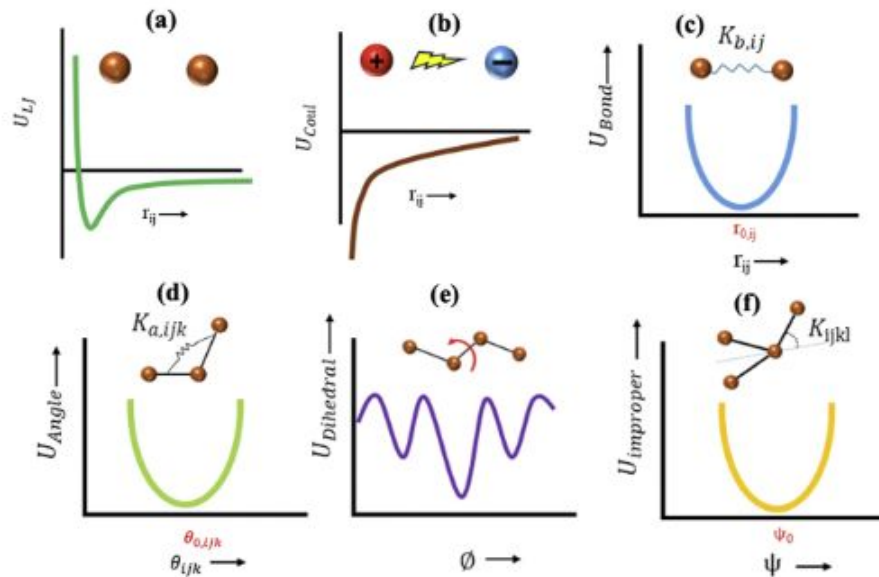
Bonded interactions

$$U_{\text{Bond}} = K_{b,ij}(r_{ij} - r_{0,ij})^2,$$

$$U_{\text{Angle}} = K_{a,ijk}(\theta_{ijk} - \theta_{0,ijk})^2,$$

$$U_{\text{Dihedral}} = \frac{1}{2}K_1[1 + \cos(\phi)] + \frac{1}{2}K_2[1 - \cos(2\phi)] \\ + \frac{1}{2}K_3[1 + \cos(3\phi)] + \frac{1}{2}K_4[1 - \cos(4\phi)],$$

$$U_{\text{improper}} = K_{ijkl}(\psi - \psi_0)^2.$$





127.0.0.1:8888/lab



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/

Name	Modified
Day1	34m ago
conda-setup-guid...	34m ago
conda-setup-guid...	34m ago
environment.yml	34m ago
Install_Wsl_Ubunt...	34m ago
README.md	34m ago
README.pdf	34m ago
setup_conda_env.sh	34m ago

Launcher



Python 3
(ipykernel)



Console



Python 3
(ipykernel)



Other



Terminal



Text File



Markdown File



Python File

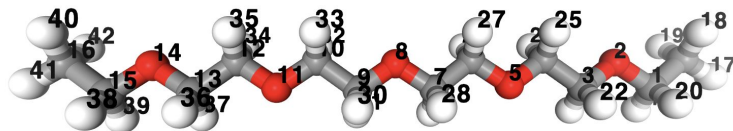


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Monomer structure



Monomer GAFF atom typing and partial charge parameters

CHARGE	0.00 (0)								
Formula: H26 C12 O5									
ATOM	1	C	PEO	1	0.931	-0.097	-0.023	-0.097100	c3
ATOM	2	C1	PEO	1	2.445	-0.109	0.007	0.132400	c3
ATOM	3	O	PEO	1	2.928	0.653	-1.094	-0.428600	os
ATOM	4	C2	PEO	1	4.355	0.677	-1.114	0.127400	c3
ATOM	5	C3	PEO	1	4.829	1.500	-2.301	0.130400	c3
ATOM	6	O1	PEO	1	6.254	1.515	-2.312	-0.430600	os
ATOM	7	C4	PEO	1	6.750	2.278	-3.409	0.130400	c3
ATOM	8	C5	PEO	1	8.269	2.261	-3.377	0.129400	c3
ATOM	9	O2	PEO	1	8.762	3.024	-4.475	-0.429600	os
ATOM	10	C6	PEO	1	10.186	3.036	-4.489	0.129400	c3
ATOM	11	C7	PEO	1	10.660	3.858	-5.676	0.130400	c3
ATOM	12	O3	PEO	1	12.084	3.872	-5.692	-0.430600	os
ATOM	13	C8	PEO	1	12.575	4.634	-6.792	0.130400	c3
ATOM	14	C9	PEO	1	14.095	4.618	-6.764	0.127400	c3
ATOM	15	O4	PEO	1	14.598	5.379	-7.860	-0.428600	os

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Jupyter Notebooks/Labs

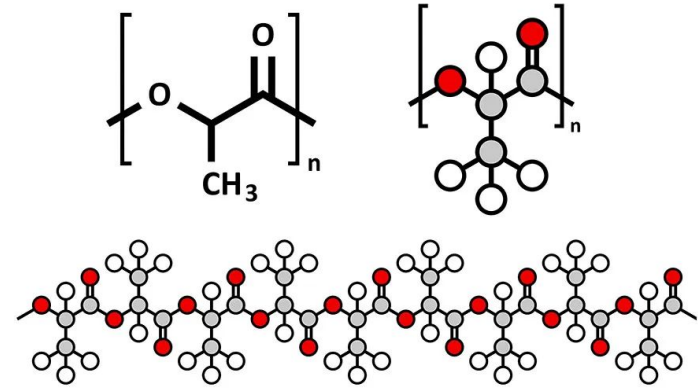
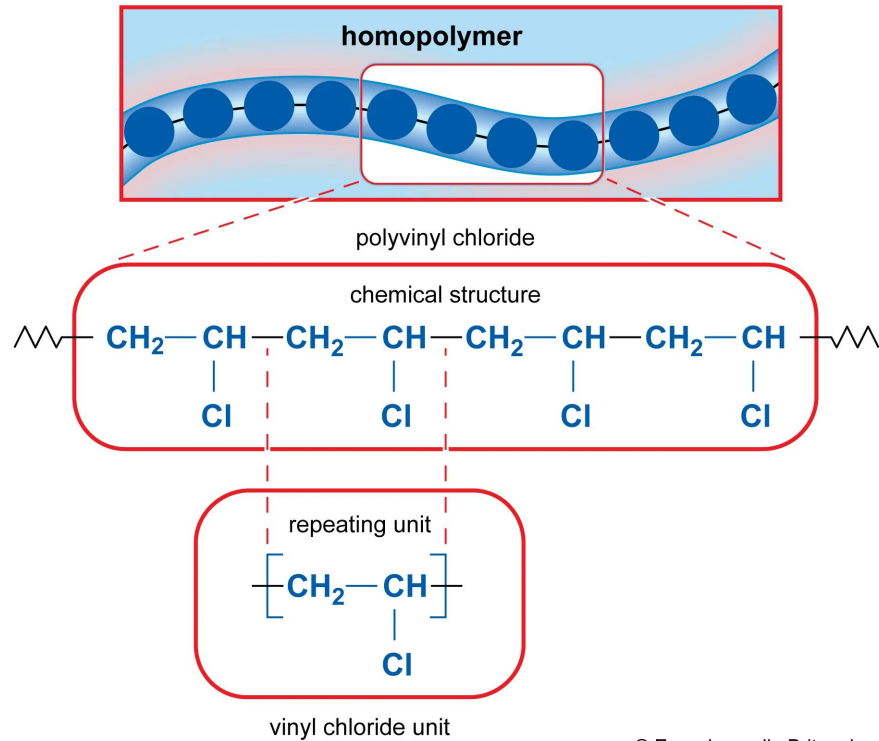
- Jupyter Notebook Basics:

<https://jupyter-notebook.readthedocs.io/en/stable/examples/Notebook/Notebook%20Basics.html>

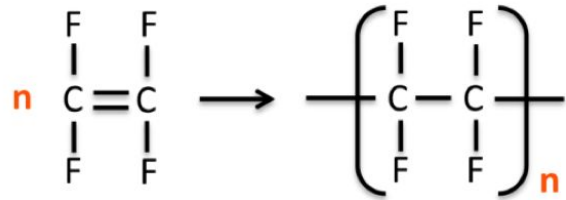
- Jupyter Lab Interface:

<https://jupyterlab.readthedocs.io/en/stable/user/interface.html>

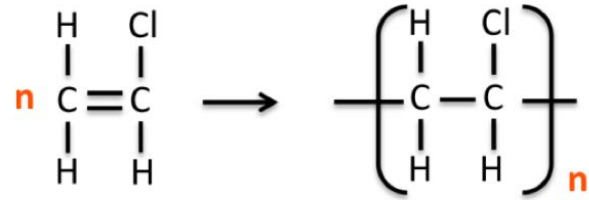
Polymer chain architecture



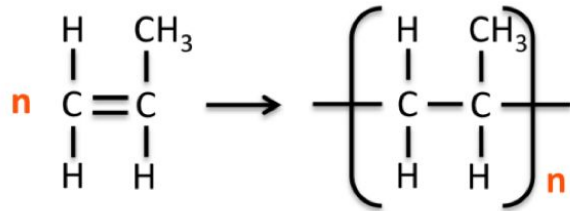
Polymer chain architecture



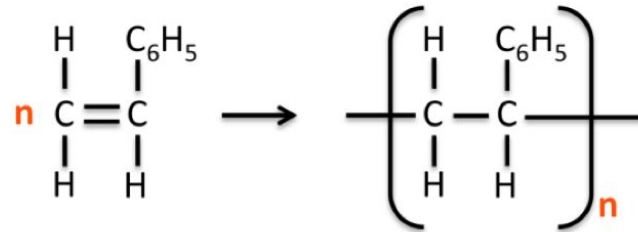
tetrafluoroethene poly(tetrafluoroethene)
PTFE



chloroethene poly(chloroethene)
vinyl chloride *PVC*

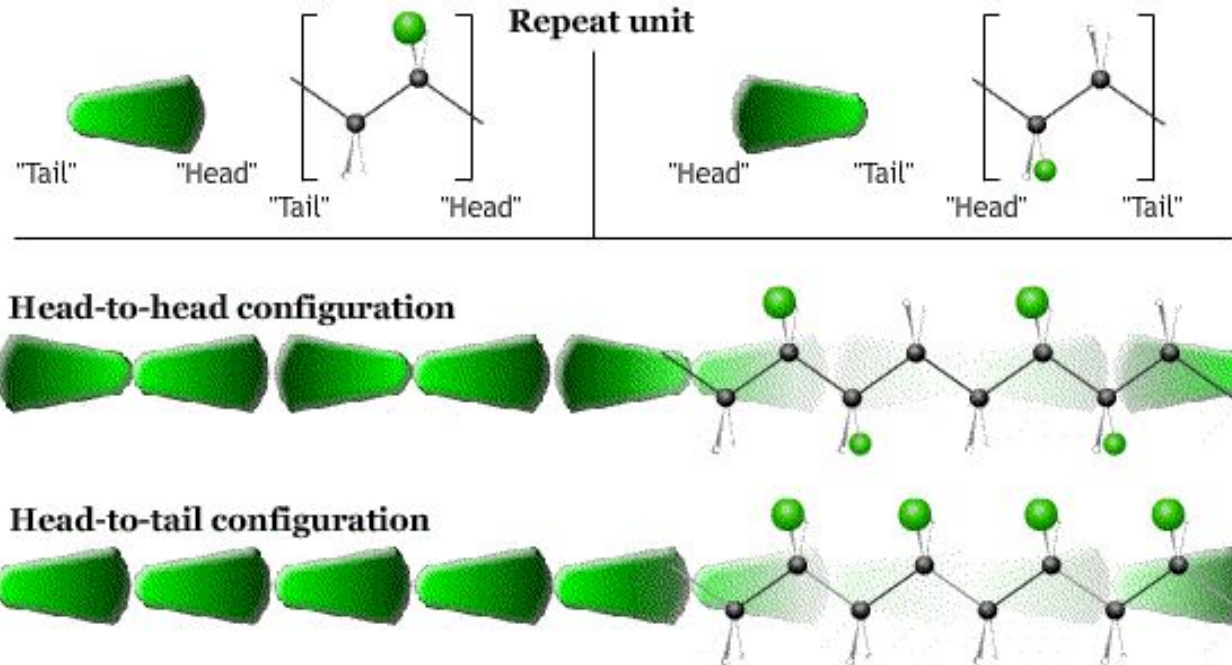


propene poly(propene)

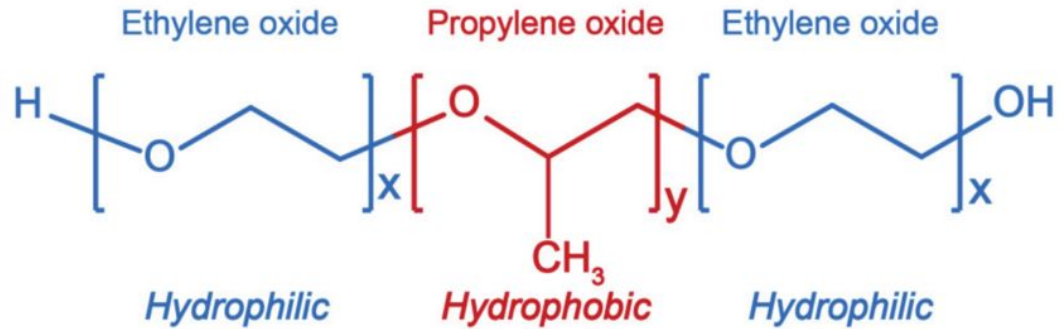


phenylethene poly(phenylethene)
styrene *polystyrene*

Polymer chain architecture

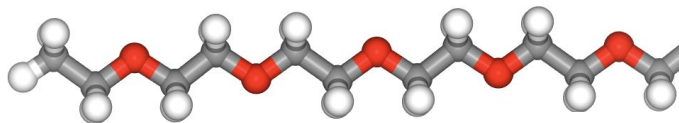
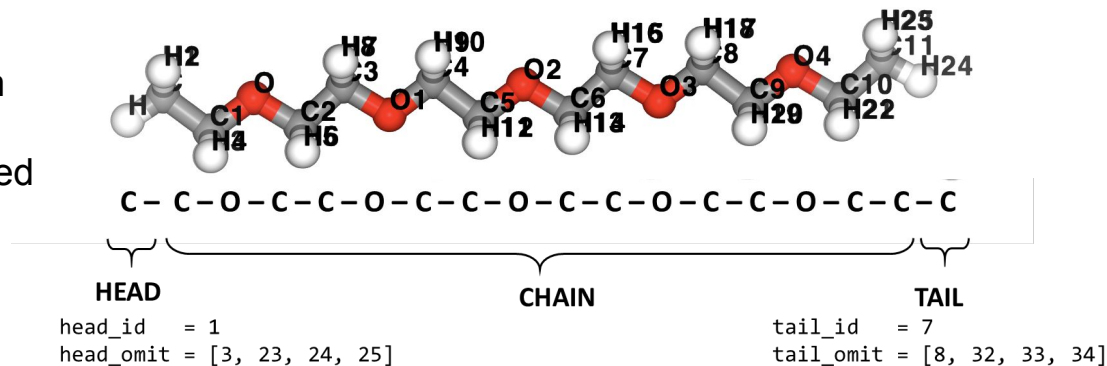


Polymer chain architecture - PEO monomers

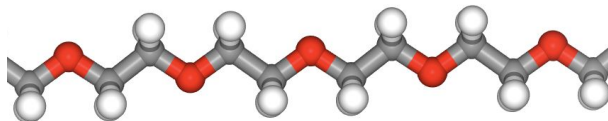


Polymer chain architecture - PEO

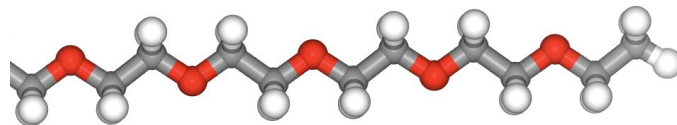
head_id is the Atom label where the Head end group is connected



Head (HTP)



Chain (PEO)



Tail (TPT)